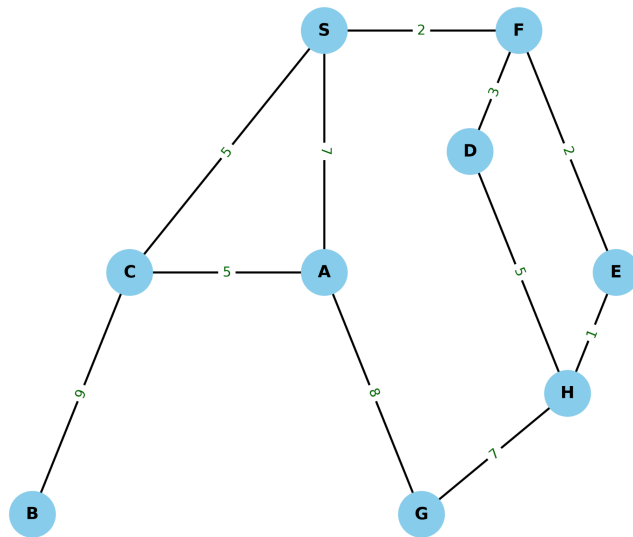


Assignment 1

1. Perform GBFS on **Graph 1** using the given $h_1(n)$ from **Table 1** and construct the shortest path from node S to node G. Show each step of the process in detail. Is your path optimal? Explain why or why not.

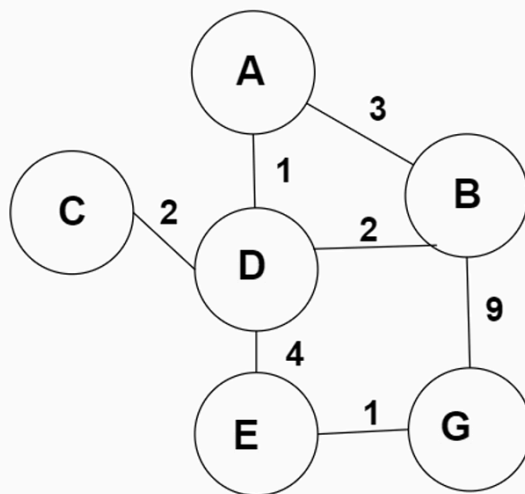


Graph 1

Heuristic Table	
Nodes	$h_1(n)$
A	14
B	16
C	10
D	8
E	15
F	9
G	0
H	17
S	12

Table 1

2. Which shortcomings of GBFS does A* Search solve? Demonstrate A* Search on the graph above.



Graph 2

Nodes,n	$h_1(n)$	$h_2(n)$
A	4	5
B	6	6
C	4	5
D	3	4
E	1	1
G	0	0

Table 2

Assignment 1

3. In the context of **Graph 2** and **Table 2**, explain which set of heuristics should work better and why.
4. Picking the better heuristic, perform an A* search on Graph 2. Show each step with detailed calculations.
5. Verify whether your chosen heuristic is i) admissible and ii) consistent or not. Show your reasoning in detail.
6. Show that, for a consistent heuristic, the evaluation function value gets non-decreasing along any possible path. Propose a new set of heuristics h_3 that would be consistent with the graph above.

Consider a maze of size $N \times N$, where horizontal and vertical moves cost 1 and diagonal movements cost $\sqrt{2}$. Let h_1 = Euclidean distance and h_2 = Manhattan distance.

7. Prove that h_2 is not admissible.
8. Show that, if h_1 is multiplied by 1.2, h_1 will not be admissible.
9. Briefly explain the drawbacks of local search. Taking an example of the 8-puzzle problem, demonstrate those drawbacks.
10. What is known as Simulated Annealing? Write down the pseudocode and explain how the concept of probability is implemented in this technique.